MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2020

Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Let f and g be the functions

$$f : [0, 2\pi] \to \mathbb{R}, \quad f(x) = \cos(x),$$

and

$$g : [0, 20] \to \mathbb{R}, \quad g(x) = x^2.$$

Determine the domain, codomain and function rule for the following compositions

(i) $f \circ g$,

(ii) $g \circ f$. (10)

(b) (i) State what it means for a function f to have a critical point at a.

(ii) Find the absolute minimum and absolute maximum of

$$f : [-3, 1] \to \mathbb{R}, \quad f(x) = x^4 - 8x^2.$$

(10)

(c) Let $P = (2, -4, -1)$ and $Q = (3, -5, 5)$ and $R = (6, -7, -2)$ be three points in $\mathbb{R}^3$.

(i) Find the distance between P and Q.

(ii) Determine, with justification, whether $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are orthogonal.

(iii) Find the angle between $\overrightarrow{PQ}$ and the vector $\vec{v} = (2, 1, -2)$. (10)

(d) Compute the following limits

(i)

$$\lim_{x \to \infty} \frac{3x^2 - 5x + 1}{7x^2 + 8},$$

(ii)

$$\lim_{x \to 0} \frac{\sqrt{2x + 1} - 1}{x}.$$

(10)

(e) Evaluate the following sums

(i)

$$\sum_{j=0}^{N} \frac{2^{(j+3)}}{5^{(j+1)}},$$

(ii)

$$\sum_{n=1}^{99} \left(\frac{1}{n^{3/2}} - \frac{1}{(n+1)^{3/2}} \right).$$

(10)

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SECTION B

2. (a) Write down the first four non-zero terms in the Maclaurin series expansion of $(3 + x^2)^{-1}$. (5)

   (b) Use the ratio test to find the radius of convergence of the power series

$$\sum_{p=1}^{\infty} p^{-p} p! x^{2p}.$$

   (8)

   (c) Let $y = \arctan(x)$. Show that

$$\frac{dy}{dx} = \frac{1}{x^2 + 1}.$$

   (4)

   (d) Write down the general form of the first three terms in the Taylor expansion of f about $x = 1$, and apply this formula to $f(x) = \arctan(x)$, giving your answer in its simplest form. (8)

   [**25**]

3. (a) Determine whether the series

$$\sum_{n=3}^{\infty} \frac{n(-1)^n}{n^2+9}$$

converges or diverges. (7)

(b) A cycloid has the parameterisation

$$x(t) = t - \sin(t) \quad y(t) = 1 - \cos(t)$$

where t is real. Show that

$$\left(\frac{dy}{dx}\right)^2 = \frac{2}{y} - 1.$$

(7)

(c) Let $\vec{x}$ be a vector in $\mathbb{R}^3$. Find all possible vectors $\vec{x}$ such that $|\vec{x}| = 1$ and

$$\vec{x} \times (1,1,1) = (1,-1,0)$$

where $a \times b$ denotes the cross product of vectors $\vec{a}$ and $\vec{b}$. (6)

(d) Sketch the curve defined in polar coordinates by $r = 1 + \sin(2\theta)$ for $0 < \theta < 2\pi$, clearly identifying the points furthest from the origin and the intersections with the coordinate axes. (5)

[**25**]

4. (a) The function f is defined for all real x by

$$f(x) = \begin{cases} e^{-1/x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

Determine, with justification, whether f is continuous. (6)

(b) Let g be a function defined for all real x by $g(x) = x^2$. Define the derivative of a function from first principles, and use it to derive the derivative of g. (6)

(c) In the following questions, you may use without proof the fact that

$$(\vec{x} \times \vec{y}) \times \vec{z} = (\vec{x} \cdot \vec{z})\vec{y} - (\vec{y} \cdot \vec{z})\vec{x}$$

for any vectors $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^3$.

(i) Suppose $\vec{a}$ is a nonzero vector in $\mathbb{R}^3$, and $\vec{b}$ and $\vec{c}$ are vectors such that

$$\vec{a} \times \vec{b} = \vec{a} \times \vec{c} \quad \text{and} \quad \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}.$$

Prove that $\vec{b} = \vec{c}$. (6)

(ii) By considering $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$, prove that for any vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d} \in \mathbb{R}^3$,

$$(\vec{b} \cdot (\vec{c} \times \vec{d}))\vec{a} - (\vec{a} \cdot (\vec{c} \times \vec{d}))\vec{b} + (\vec{d} \cdot (\vec{a} \times \vec{b}))\vec{c} - (\vec{c} \cdot (\vec{a} \times \vec{b}))\vec{d} = 0.$$

(7)

[**25**]